Chapter

1

Chapter Learning Outcomes

CLO

1

- Define the meaning of Intelligence.
- Classify AI systems.
- Distinguish between strong and weak AI.
- Know and understand the different definitions of AI.
- Describe the Turing test, its purpose, and its limitations.
- Understanding of the historical evolution of Artificial Intelligence.
- Define the meaning of expert systems as well as their main components and know some early expert systems.
- List the main AI sub-disciplines.
- Know and list the best AI programming languages in the last year.

What is Artificial Intelligence?

It is difficult to give a precise definition of artificial intelligence. Some recent artificial intelligence scientists have attempted to define artificial intelligence (in short AI) in various ways.

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What is Artificial Intelligence?

A more difficult question is: What is intelligence?

- **Intelligence means:**
  - **Ability to learn.**
  - **Ability to understand.**
  - **Ability reason/think.**
  - **Ability to solve problems.**
  - **Efficiently at solving problems.**
  - **Innovation at solving problems.**

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What is Artificial Intelligence?

Can computers become intelligent?

Since human being live once time, Artificial intelligent (A.I) tries to transfer the experience and knowledge of human being to computers. This is Artificial intelligent.

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What is Artificial Intelligence?

There are four categories for the definitions of AI based on whether an agent thinks/acts humanly/rationally.

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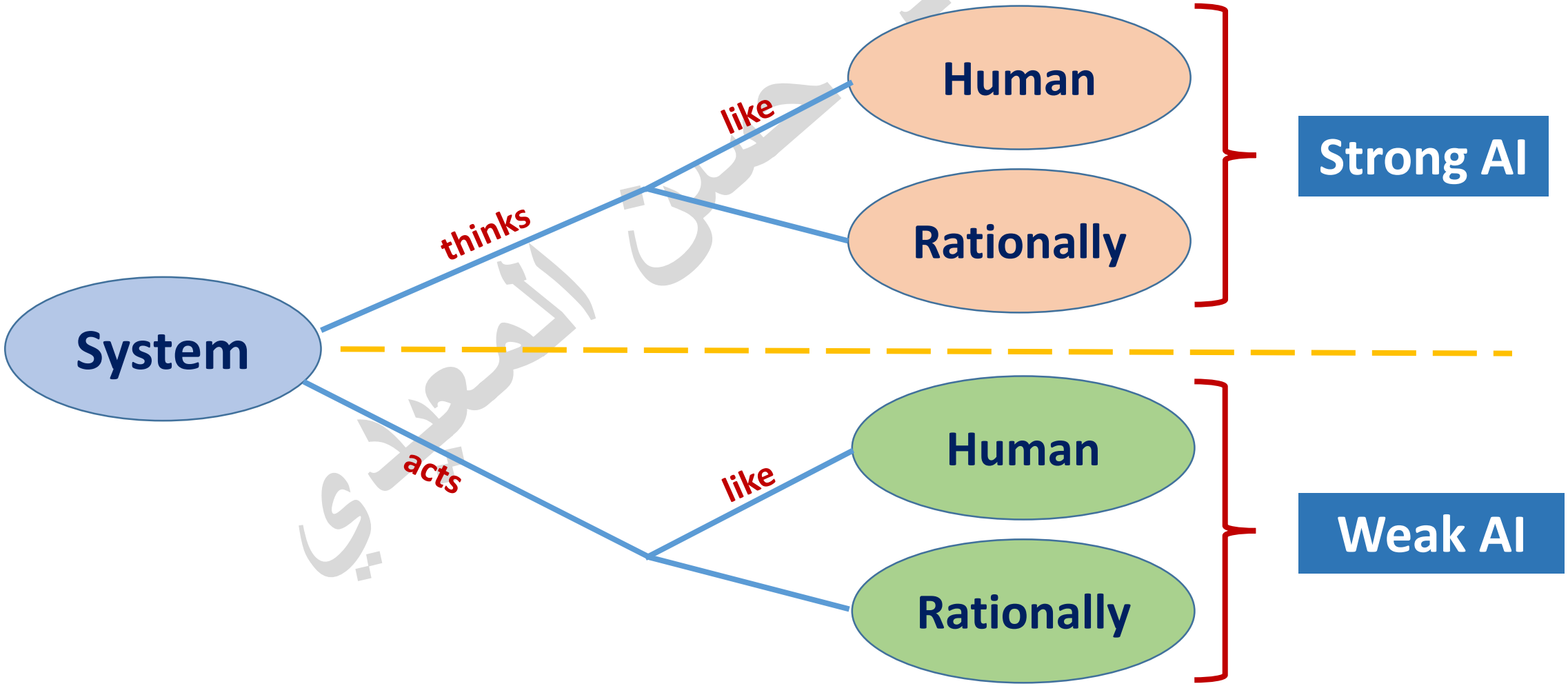
What is Artificial Intelligence?

A general classification of AI systems, due to Russell and Norvig



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Major Branches of AI

Weak AI

The study and design of machines that perform intelligent tasks. Not concerned with how tasks are performed, mostly concerned with performance and efficiency.

- E.g., to make a flying machine, use logic and physics, don't mimic a bird.

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Major Branches of AI

Strong AI

The study and design of machines that simulate the human mind to perform intelligent tasks. Borrow many ideas from psychology, neuroscience. Goal is to perform tasks the way a human might do them – which makes sense, since we do have models of human thought and problem solving.

- **This course is concerned with weak AI.**

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Major Branches of AI

Strong AI

- Duplication of intelligence
- Aims to understand intelligence

Weak AI

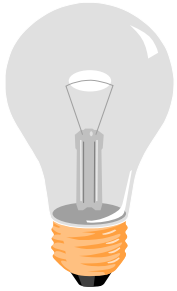
- Simulation of intelligence
- Aims to make computers more useful

What is Artificial Intelligence?

Thinking Humanly	Thinking Rationally
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Acting Humanly	**Acting Rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998) "AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

What is Artificial Intelligence?



AI is the branch of computer science concerned with making computers behave like humans.
(Russell and Norvig, 2011)

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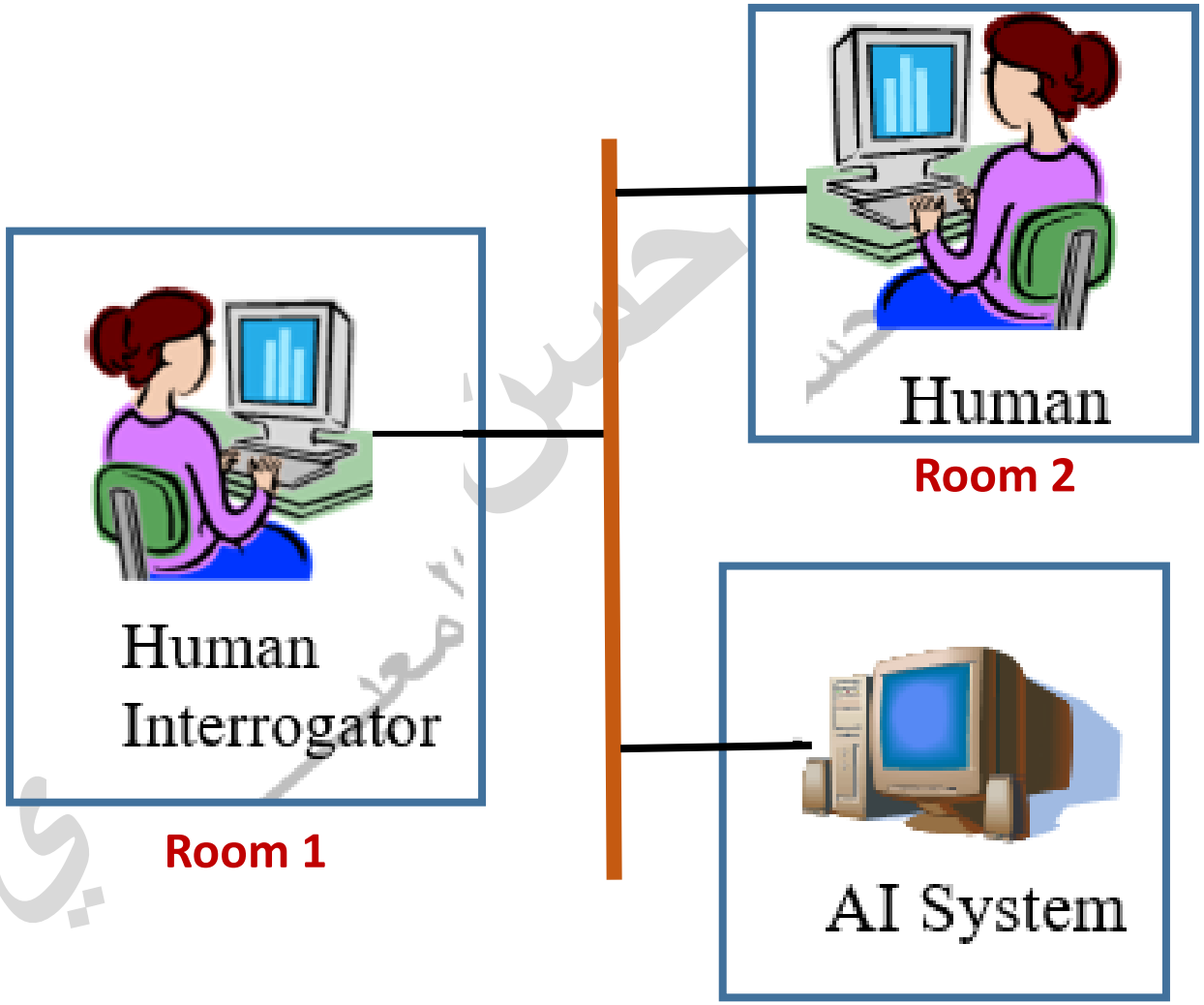
Turing Test

Alan Turing 1950



Turing Test

- A human interrogator sits in one room and uses a computer to communicate with respondents A and B in another room.
  - One of A and B is a human, the other is a computer.
  - The interrogator does not know which is a computer.
- The interrogator tries to determine which is the human and which is the computer.
- The machine tries to fool the interrogator into believing that it is the person.

- **If the computer can fool enough humans interrogators, it is considered intelligent** .

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Birth of AI

1956 Dartmouth AI Project

Dartmouth workshop 1956, organized by Alan Turing, John McCarthy, Marvin Minsky, Allen Newell.
Considered the birth of AI.

John McCarthy proposed the name ARTIFICIAL INTELLIGENCE

History of AI

1943	Walter Pitts and Warren McCullock showed how artificial neural networks could compute, relying on the use of feedback loops
1950	Turing proposed his test, the Turing test, to recognize machine intelligence.
1956	John McCarthy proposed the name "Artificial Intelligence" at Dartmouth University.
1958	John McCarthy developed the LISP program at MIT for AI work.
1963	The Stanford University founded the Artificial Intelligence Laboratory under John McCarthy
1965	The Robotics Institute was started at Carnegie Mellon University under Raj Reddy.
1965	Joseph Weizenbaum (MIT) built ELIZA, an interactive program that carries on a dialogue in English on any topic.

History of AI

1965 to 1975	Edward Feigenbaum and Robert K. Lindsay at Stanford built DENDRAL, the first expert system
1969	A mobile robot called Shakey was assembled at Stanford.
Early 1970s	First practical demonstration of the use of fuzzy logic
1971-1972	Alain Colmerauer and Phillipe Roussel wrote the computer language, PROLOG
1972	Edward Shortliffe, a Stanford doctoral student under Bruce Buchanan, and others wrote MYCIN, an expert system to diagnose infectious blood diseases and recommend antibiotics
Late 1970s	First commercial expert system was developed. It was XCON (for eXpert CONfigurer), developed by John McDermott at Carnegie Mellon
1986	Rise of Machine Learning
1997	Deep Blue

The State of the Art

1. Game Playing
2. Automated Reasoning and Theorem Proving
3. Expert Systems
4. Natural Language Understanding
5. Vision
6. Planning and Robotics
7. Machine Learning
8. Neural Nets and Genetic Algorithms

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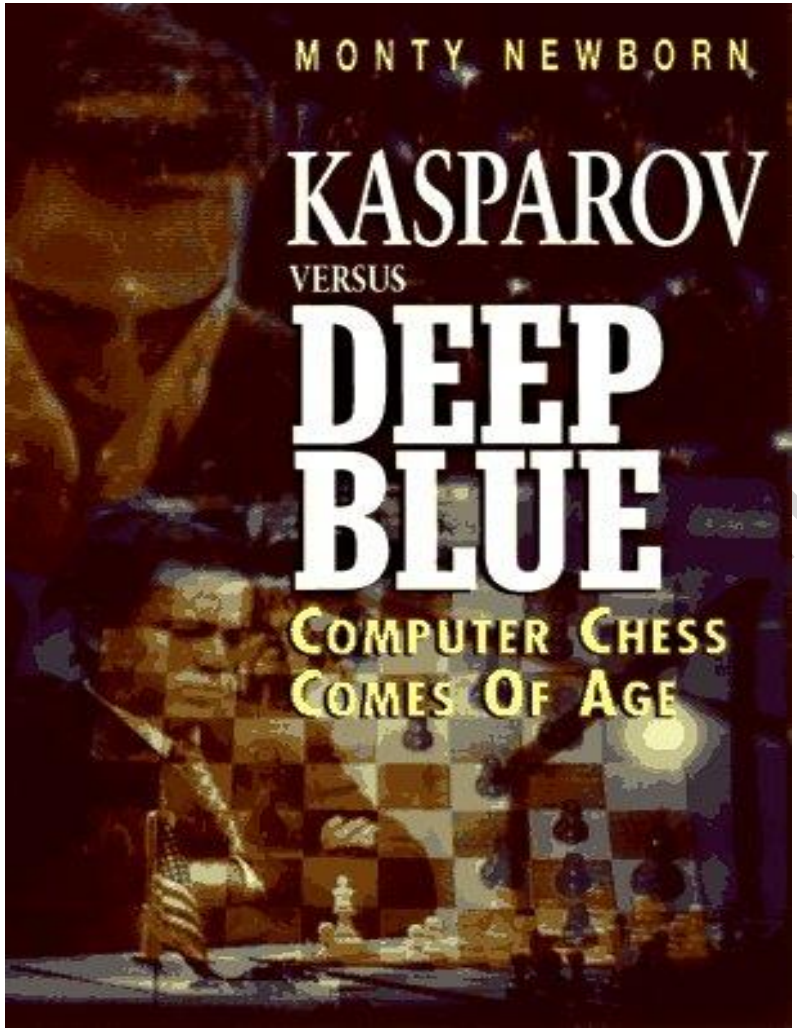
Game Playing



Kasparov versus Deep Blue

Kasparov versus Deep Blue

- **First Match**
  - February 10, 1996: took place in Philadelphia, Pennsylvania
  - Result: **Kasparov**–Deep Blue (4–2)

The 1996 match

Game #	White	Black	Result	Comment
1	**Deep Blue**	Kasparov	1–0	
2	**Kasparov**	Deep Blue	1–0	
3	Deep Blue	Kasparov	½–½	Draw by mutual agreement
4	Kasparov	Deep Blue	½–½	Draw by mutual agreement
5	Deep Blue	**Kasparov**	0–1	Kasparov offered a draw after the 23rd move.
6	**Kasparov**	Deep Blue	1–0	
***Result:* Kasparov–Deep Blue: 4–2**				

Kasparov versus Deep Blue

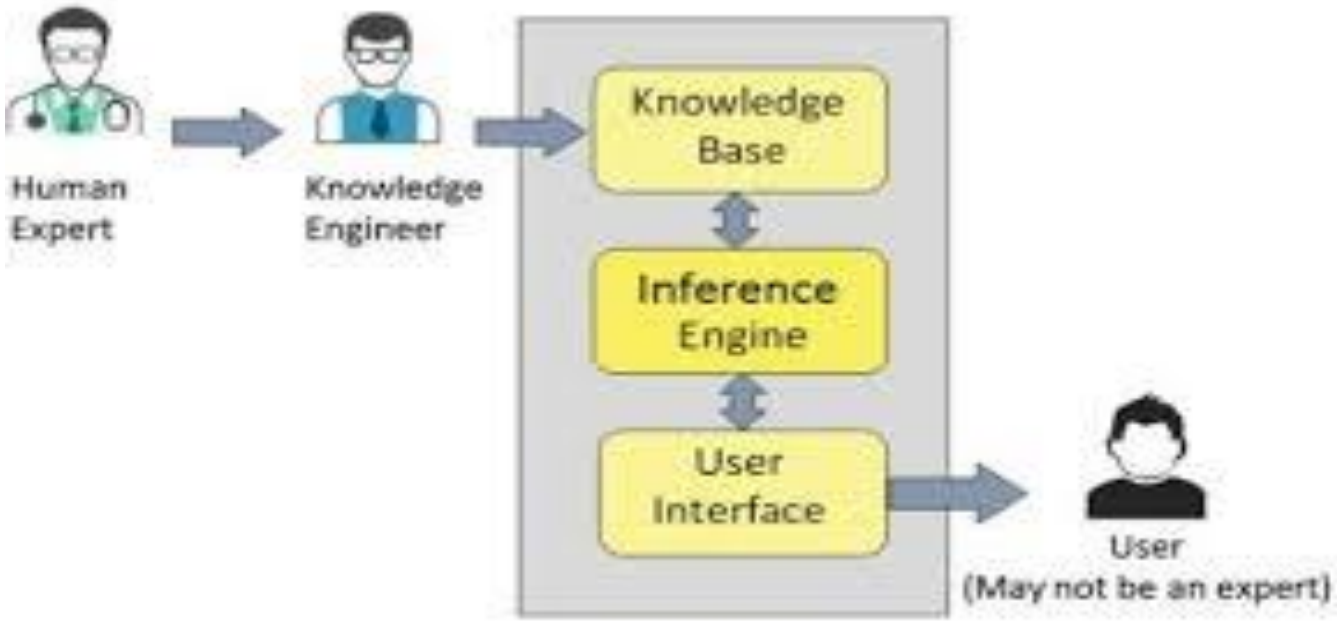
- **Second Match**
  - May 11, 1997: held in New York City, New York
  - Result: Deep Blue–Kasparov (3½–2½)

The 1997 rematch

Game #	White	Black	Result	Comment
1	**Kasparov**	Deep Blue	1–0	
2	**Deep Blue**	Kasparov	1–0	
3	Kasparov	Deep Blue	½–½	Draw by mutual agreement
4	Deep Blue	Kasparov	½–½	Draw by mutual agreement
5	Kasparov	Deep Blue	½–½	Draw by mutual agreement
6	**Deep Blue**	Kasparov	1–0	
***Result:* Deep Blue–Kasparov: 3½–2½**				

Expert Systems

Expert system is a rule based computer system which simulates a human expert in his/her field of expertise in an attempt to solve a particular problem.



Expert Systems Programming Languages

CLIPS

JESS

PROLOG

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Early Expert Systems

- **DENDRAL**

  Used to identify the structure of chemical compounds. First used in 1965

- **MYCIN**

  Medical system for diagnosing blood disorders. First used in 1979

- **DIPMETER**

  Geological data analysis for oil

- **PROSPECTOR**

  Geological data analysis for minerals

- **XCON/R1**

  Configuring computer systems

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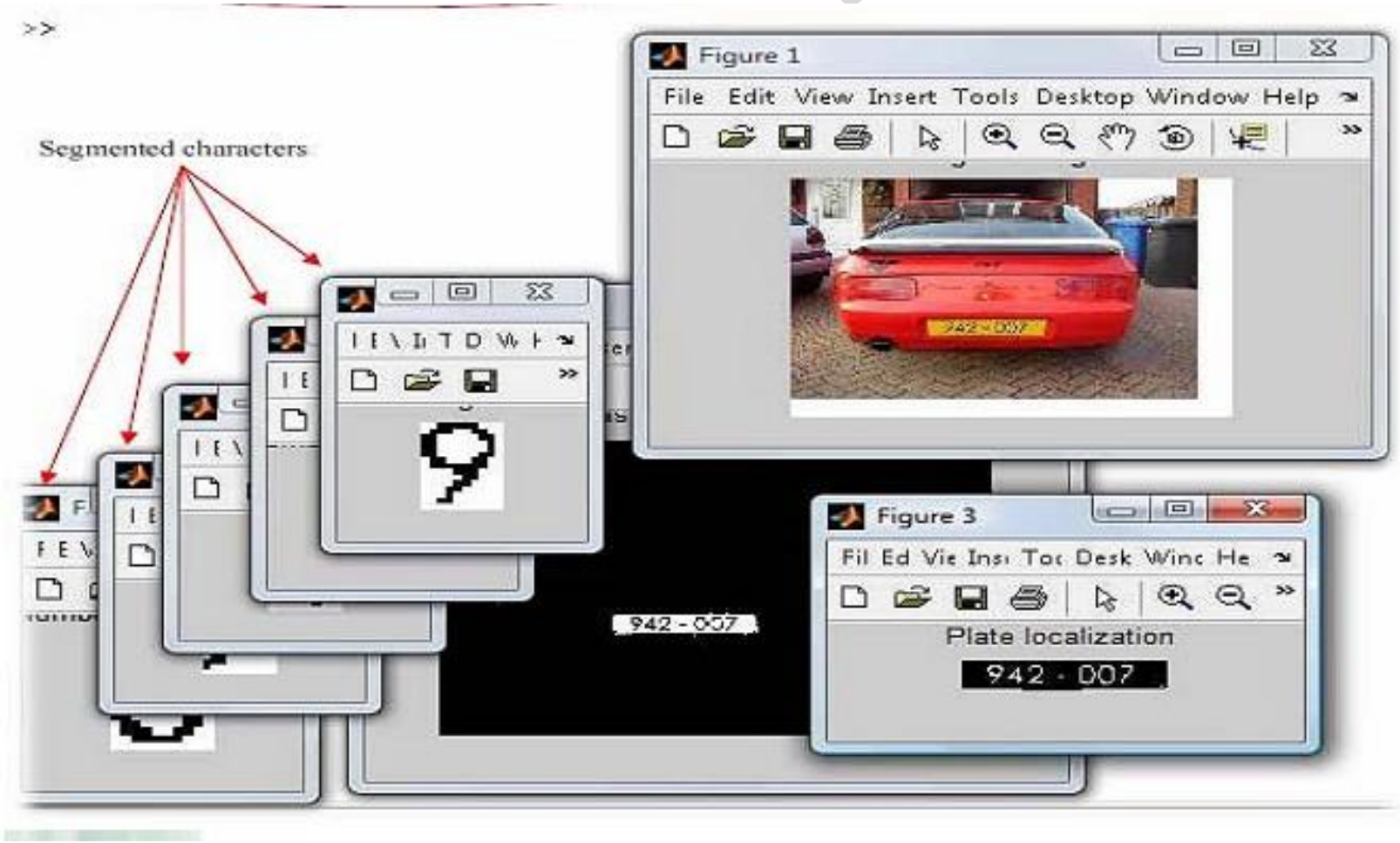
Computer Vision

Computer Vision is an AI field that deals with how computers can be made for gaining high-level understanding from digital images or videos.

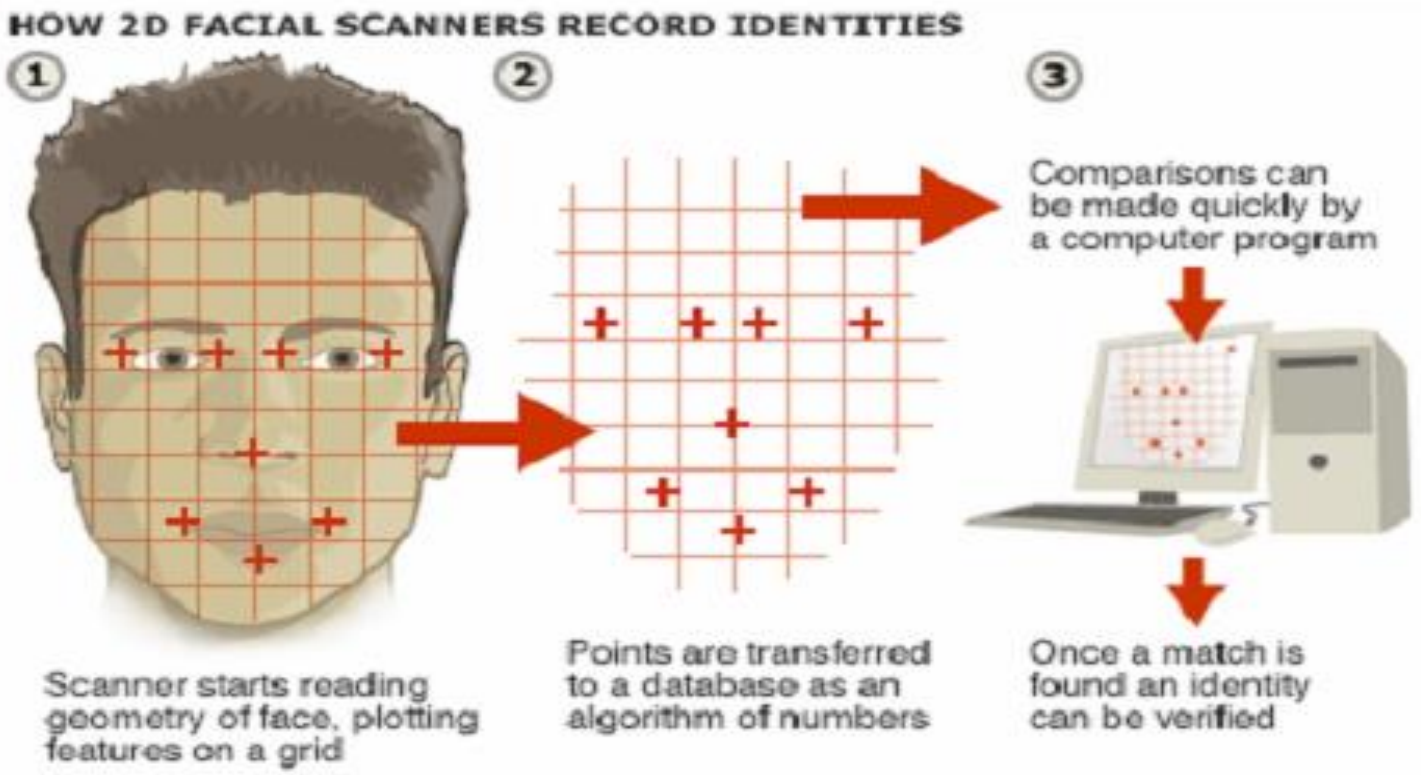
Computer Vision is concerned with the automatic extraction, analysis and understanding of useful information from a single image or a sequence of images.

- Computer vision:

Plate Extraction



Face Recognition



Face Recognition

Robotics

Robotics - Camro

It is a robot manufactured by a Japanese company, used primarily for the care of the elderly.

Robotics - Medical Applications

Robotics - Medical Applications

Robotics - Medical Applications

Smart Homes

BRADFORD
Making Knowledge Valuable
REMOTE HOME CONTROL
ONLINE HOME AUTOMATION
CLIMATE
LIGHTING
SECURITY
ENTERTAINMENT
22°C
ECO MODE
NORMAL

Smart Cars

Early AI Programming Languages

IPL

ROSIE

PROLOG

RITA

LISP

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8 Best AI Programming Languages in 2020

https://www.rankred.com/best-artificial-intelligence-programming-language/

- Python
- C++
- Lisp
- Java
- Prolog
- JavaScript
- Haskell
- Julia

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Check the Chapter Learning Outcomes

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End of Chapter

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